

Agentic Linguistics Analyzer (PMDD)

Design & Structure of the Final Scientific Analysis Report

1. Report Purpose & Scientific Rigor

This document outlines the structure of the final analysis report generated by the PMDD (Pragmatic Meaning Drift Detector) application. Designed strictly for the academic and scientific research community, this report acts as a verifiable scholarly artifact.

By utilizing a 5-Agent architecture with a Reflective Brain (OpenAI API powered self-correction) and Episodic Memory (learning from previous analyses), the system dynamically adapts to the corpus. Instead of strict NLP rules, it employs established Linguistic Theories to analyze large corpora, providing robust conclusions.

2. Structure of the Final Analysis Report

The final report will be formatted into the following standardized sections:

1. Executive Summary: High-level overview of meaning drift scores and core semantic shifts.
2. Methodological & Theoretical Framework: Explicit statement of linguistic theories applied (e.g., Searle, Grice, Halliday) to ensure reproducibility.
3. Quantitative Landscape (Agent 4): Statistical distributions, TTR, and keyness.
4. Qualitative Pragmatic & Semantic Analysis (Agents 2 & 3): In-depth review of speech acts, maxims, and semantic fields.
5. Interpretative Synthesis (Agent 5): How the qualitative and quantitative data converge to prove the linguistic drift.
6. Evidence Appendix: Raw data mapping with segment citations.

3. Presentation of Linguistic Evidence (30/70 Ratio)

The report balances scientific metrics with deep contextual analysis, maintaining a strict evidence ratio:

Quantitative Evidence (30%):

- Statistical Tables: Word frequencies, Collocation strength (Mutual Information), and Log-Likelihood scores.
- Semantic Drift Charts: Numerical representation of shifts in conceptual clustering.
- Example presentation: 'Keyword X showed an MI score of 6.5 in Period A, dropping to 2.1 in Period B.'

Qualitative Evidence (70%):

- Verbatim Corpus Extracts: Exact sentences pulled from the text (referenced by Segment ID) to illustrate the statistics.
- Contextual Interpretation: Explaining **why** the shift occurred using pragmatic maxims.
- Example presentation: [Segment #1042]: 'We must fight this policy.' -> Analyzed as an explicit Directive speech act, demonstrating the statistical shift toward authoritarian register.

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4. Multi-Agent Interpretation & Theoretical Support

The Orchestrator (Agent 5) synthesizes outputs from the preceding agents. Rather than presenting isolated metrics, the report triangulates findings:

A. Error-Correction (The Brain): The report highlights where the Reflective Loop corrected initial misclassifications (e.g., 'Agent 2 initially flagged Segment #45 as Assertive, but cross-referencing Agent 4's collocation data triggered a self-correction to Indirect Directive').

B. Episodic Memory Learning: The report notes how previous analyses informed current classifications, demonstrating adaptive learning rather than static NLP rules.

C. Theoretical Grounding: Every claim is tethered to theory. For example, if a shift in formality is detected, it is explicitly justified using Halliday's Register Theory (Field, Tenor, Mode). If evasion is detected, it is proven using Grice's Maxim of Quantity violations. This ensures the output is peer-review ready for the research community.